

Jessica Oldov

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EDUCATION

Duke University

Bachelor of Science in Computer Science, Minor in Russian Culture and Language

Honors: QuestBridge Scholar, Allan Kornberg Scholarship Recipient

Aug. 2021 – May 2025

GPA: 4.00/4.00

EXPERIENCE

Software Engineer

Liberty Mutual Insurance

July 2025 – Present

New York, NY

- Built a **Java 21/Spring Boot** microservice integrating **120+ risk-scoring features** across **Kafka, MongoDB, and REST APIs** for a production pricing model, reducing prediction variance by **7%** and improving annual accuracy by **\$12M+**
- Led implementation of Teen-Driver Discount model logic to production (June 2026, 3 states), scoring **10K+** eligible policies/month and reducing qualifying premiums by **8%**
- Built automated test suites (**JUnit 5, Mockito, Spring Boot Test**) validating model output correctness pre-deployment, maintaining **99%** coverage on core rating logic

Software Engineering Fellow, ML Infrastructure

Meta (PyTorch)

September 2024 – December 2024

Remote

- Optimized **PyTorch** core checkpoint serialization (**Python, C++**) for large-scale models (**1B+ parameters**), refactoring tensor storage allocation to cut serialization time by **15%**
- Contributed **12 merged PRs** to PyTorch's open-source ML infrastructure, working directly in core tensor storage and checkpointing internals used across the deep learning community
- Led sprint planning and issue triage for a 4-person contributor team, accelerating critical bug resolution by **25%**

Software Engineer Intern

Liberty Mutual Insurance

June 2024 – August 2024

Indianapolis, IN

- Built a **TypeScript/Node.js serverless payment workflow** (**AWS Lambda, API Gateway, DynamoDB**) and conversational payment flows via **Amazon Lex**, automating **500+** daily transactions and deflecting **35%** of calls from live agents

PROJECTS AND RESEARCH

Tokenizer Efficiency Across Languages | *Python, HuggingFace Transformers, tiktoken, FLORES-200*

- Benchmarked **6 tokenizers** (GPT-4, GPT-4o, Llama-3, Mistral, mBERT, NLLB-200) across **5 languages** using **FLORES-200** parallel corpus data, measuring token-efficiency disparities
- Found Mongolian required up to **3.7x** more tokens than English across general-purpose tokenizers, vs. **1.4x** for NLLB-200 - evidence that multilingual token inefficiency is a solvable design choice, not an inherent limitation

Small Claims Insurance Application | *Python, AWS Lambda, DynamoDB, Datadog, OpenAI API*

- Architected and built a **Python/AWS serverless claims platform** integrating the **OpenAI API** for conversational claim intake, increasing completion rates and reducing submission time from **12 to 4 minutes**
- Deployed the platform to **200+ users** within its first month and reduced API response latency by **30%** by optimizing **DynamoDB query patterns** and mitigating **AWS Lambda cold starts**

Classical ML vs. Deep Learning for Text Classification

Co-authored Research Paper

- Benchmarked **Random Forest, SVM, KNN** against fine-tuned **BERT** on Stanford Sentiment Treebank (SST2) using **HuggingFace** pipelines and **K-Fold CV**; BERT outperformed classical baselines on F1 (**0.95** vs. **0.87**) at higher compute cost

TECHNICAL SKILLS

Data & ML: PyTorch, TensorFlow, HuggingFace, OpenAI API, scikit-learn, pandas, NumPy

Languages: Java, Python, TypeScript, JavaScript, C++, SQL

Frameworks & Libraries: Spring Boot, React, Node.js, Next.js, Express.js

Cloud & Infrastructure: AWS (Lambda, DynamoDB, API Gateway, S3, SQS), Kafka, Docker, PostgreSQL, MongoDB, Redis, Git, CI/CD